# PAVAN KUMAR ETTA

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## EDUCATION

**University of Massachusetts, Dartmouth | GPA: 4.0** — **North Dartmouth, USA**
*M.S. in Computer Information Science, Data Science* — *Aug 2024 - May 2026*
**Indian Institute of Information Technology | GPA: 3.2** — **SriCity, INDIA**
*B.Tech in Electronics and Communication Engineering* — *Aug 2017 - Jun 2021*

## SKILLS

**Programming:** Python, JavaScript, TypeScript, C/C++, Java, Go
**Software Engineering:** Data Structures & Algorithms, OOP, Microservices, Distributed Systems
**Backend & APIs:** Django, Flask, FastAPI, REST APIs, gRPC, GraphQL, Event-Driven Architecture
**Frameworks:** jQuery, Angular, Node.js, Vue.js, Next.js, Flutter, Nginx, Figma, HTTPS
**Databases & Caching:** PostgreSQL, MySQL, DynamoDB, MongoDB, Redis, Elasticsearch
**Cloud & Infra:** AWS (Lambda, EC2, S3, RDS, MSK, EKS), GCP, Azure, Terraform, Serverless
**Data Engineering:** Apache Spark, PySpark, Airflow, Snowflake, Hadoop, dbt
**DevOps & Observability:** Docker, Kubernetes, CI/CD, GitHub Actions, GitLab CI

## WORK EXPERIENCE

**Graduate Research Assistant | University of Massachusetts, Dartmouth | USA** — **Mar 2025 – Dec 2025**

- Led development of a financial-reporting platform using **Apache Spark**, **Hive**, and **Hadoop** to process uploaded documents, migrated report-generation logic from Java to Scala, and built a React.js validation interface for verifying NLP-generated reports.
- Integrated LangChain for AI workflow orchestration and optimized ChatGPT-based text extraction, improving extraction accuracy by **30%**, implemented BERT-based classification with **93%** accuracy.
- Designed and implemented a scalable backend pipeline using **Flask,** MongoDB, Redis, and **Docker** to orchestrate document uploads, asynchronous report generation, validation, and persistent storage, improving system reliability and maintainability.

**Software Engineer II | Rhenix Lifesciences LLP | India** — **Mar 2022 – Jul 2024**

- Engineered CureOrg with integrations across 5+ wearable platforms using **OAuth 2.0** and REST APIs, enabling synchronization of health data for **100+ users** and processing **1M+ health records**, including activity, sleep, and heart-rate metrics.
- Automated daily health-data processing with Python jobs and **Django** management commands on **AWS EC2**, reducing manual collection by **90%** through token refresh, JSON parsing, retries, validation, and MySQL persistence.
- Improved reliability of cross-functional health-data services using **Kubernetes** orchestration and optimized **GraphQL** queries, while implementing audit logs and alerts that reduced synchronization debugging time by **40%** and accelerated failure detection.
- Built a scalable data-access layer using indexed MySQL queries, **Redis caching**, and paginated **REST endpoints**, improving health-metrics retrieval performance by **70%** for frequently accessed user data.
- Established unit, integration, and regression testing for Go and Python services, validating **PostgreSQL** workflows and Dockerized deployments across **ECS**, **S3**, and **SQS** while improving defect detection, maintainability, and release confidence.
- Owned cross-platform feature delivery across backend, Android, iOS, QA, and DevOps teams by translating customer requirements into technical tasks, resolving release blockers, and supporting Google Play and App Store deployments.

**Software Engineer Intern | Cure Science Institute | USA** — **Jun 2021 – Feb 2022**

- Developed a **BLE-based** asset tracking system using an Android app, BLE beacons, gateways, and REST APIs, enabling real-time location data uploads to AWS S3 with persistent background services for **24/7** tracking and improved battery efficiency.
- Implemented **Go-based** monitoring services with health checks, structured logging, and configuring **New Relic** alerts and Slack notifications to detect offline gateways, API failures, and delayed updates, improving production visibility and incident response.
- Engineered a scalable backend using **AWS Lambda,** API Gateway, DynamoDB, RDS, and scheduled jobs to process device telemetry, persist location data, and support real-time asset monitoring and analytics.

## PROJECTS

**Distributed API Gateway & Rate Limiting Service** [GITHUB] — ***(Python, Go, AWS, Redis, Docker, Kubernetes)***

- Designed a cloud-native API gateway with authentication, request routing, rate limiting, retries, circuit breakers, and service-level observability, improving backend reliability and protecting microservices from traffic spikes.

**Fault-Tolerant Sharded Key-Value Store** [GITHUB] — ***(C++, Raft, SQLite)***

- Engineered a fault-tolerant sharded key-value store in **C++** using Raft consensus, SQLite, leader election, log replication, and replicated state machines to maintain consistency and support recovery from server failures.

**Automated Subjective Answer Evaluation System** [GITHUB] — ***(Flask, BERT, Machine Learning, Docker, AWS EC2/S3)***

- Developed an automated answer-evaluation platform using Python, Flask APIs, BERT, Decision Trees, Random Forest, and XGBoost to process more than 10,000 responses and achieve **95%** classification accuracy.

**Wandermate** [GITHUB] — ***(React, TypeScript, Node.js, MongoDB, Socket.IO)***

- Led the development of a full-stack travel platform using TypeScript, Node.js, and MongoDB, integrating Socket.IO for real-time messaging, PropelAuth for secure authentication, Mapbox for interactive maps, and reusable APIs for third-party integrations.